SCX λ $\lambda < 0$

Status

SCX λ Directionality and the $\lambda < 0$ Divergent Case

Condition Formalization, Terminal States, and Early Warning Signals

Equality Principleawaiting

Directional Analysis of Inequality Dynamics within the Equality Principle Framework

SCX

July 2, 2026

Abstract

Abstract SCXEquality PrincipleCoreawaiting $\Delta(t)$ whereGrowth λ $\lambda \geq 0$ textbf$\lambda < 0$ ——i.e.awaiting $\lambda < 0$ iff condition **ITA**Intergenerational Trauma Amplification **IPI**Identity Politics Industrialization **SR**Segregation Reinforcementeachdefine—— we derive $\Delta(t) = \Delta_0 e^{|\lambda|t}$ StatusFragmentationSR RevolutionIPIExtinctionITA **CEWI**

English Abstract: The core dynamical equation of SCX Equality Principle describes the continuous-time evolution of the inequality measure $\Delta(t)$, where the sign of the growth-rate parameter λ determines the system's qualitative behavior. Prior work has focused primarily on the $\lambda \geq 0$ (convergent or steady-state) regime. This paper systematically formalizes the textbf$\lambda < 0$ divergent regime—in which inequality intensifies at an exponential rate. We identify three necessary-and-sufficient conditions that trigger $\lambda < 0$: **ITA** (Intergenerational Trauma Amplification), **IPI** (Identity Politics Industrialization), and **SR** (Segregation Reinforcement). Each condition is rigorously formalized through the definition—mechanism—activation-function triad. On this basis, we derive the divergent dynamics $\Delta(t) = \Delta_0 e^{|\lambda|t}$ with stochastic bounds, establish the bifurcation conditions for three terminal states: **Fragmentation** (SR-dominant), **Revolution** (IPI-dominant), and **Extinction** (ITA-dominant). We further construct five early warning signals and synthesize them into the **Composite Early Warning Index (CEWI)**, and finally provide analytical formulas for intervention pathways and optimal intervention timing. The entire exposition is rigorously formalized in Chinese and English.

Contents

1 λ

2 Introduction: The Problem of λ Directionality

2.1 SCX awaiting / Review of SCX Inequality Dynamics

SCX Equality Principle awaiting $\Delta(t)$

In the SCX Equality Principle framework, the continuous-time evolution of the inequality measure $\Delta(t)$ is governed by the following stochastic differential equation:

Definition 2.1 (SCX awaiting / SCX Inequality Dynamics Equation)**.** $\Delta(t) \in [0, \infty)$ *t awaiting $\lambda \in \mathbb{R}$ Growth*

$$d\Delta(t) = \lambda\,\Delta(t)\,dt + \sigma\,\Delta(t)\,dW_t,$$

where $\sigma > 0$ W_t

λ awaiting

- $\lambda > 0$Convergent regime—— awaitingDecaysawaiting
- $\lambda = 0$Critical regime—— awaiting
- $\lambda < 0$Divergent regime—— awaiting

Proposition 2.2 (/ Itô Solution)**.** *define 2.1 SDE*

$$\Delta(t) = \Delta_0 \exp\left(\left(\lambda - \frac{\sigma^2}{2}\right)t + \sigma W_t\right),$$

where $\Delta_0 = \Delta(0)$ $\lambda < 0$ $|\lambda| > \sigma^2/2$ $|\lambda| - \sigma^2/2 > 0$

2.2 / Limitations of Prior Work

Prior work in the SCX framework has predominantly examined the $\lambda \geq 0$ regime, focusing on convergence conditions (Thm4, Thm5 of the SCX axiom system) and the mechanisms—such as κ-suppression (see Civilization Gauge paper)—that can mask or delay the consequences of $\lambda < 0$. However, three critical gaps remain:

$\lambda \geq 0$ SCXAxiom SystemThm4Thm5 $\lambda < 0$ κawaitinghowever

1. $\lambda < 0$ λ **The *causes* of $\lambda < 0$ are unformalized:** What structural social conditions necessarily produce negative λ? This is a structural-attribution problem, not a parameter-fitting problem.
2. **Classification:** $\lambda < 0$ — **Divergent endpoints are unclassified:** $\lambda < 0$ guarantees collapse, but the *form* of collapse (fragmentation, revolution, or extinction?) depends on which condition dominates.
3. there exists——for exampleACFsimultaneously **Early warning lacks a composite index:** Five independent signals can conflict—e.g., rising ACF with falling variance. A framework to synthesize them into a single actionable index is needed.

This paper fills all three gaps.

3 λ define

4 Formal Definition of λ and Notational Conventions

4.1 λ / Micro-Foundations of λ

λ

λ is not an exogenously given parameter but an emergent quantity from the internal structural features of the social system:

Definition 4.1 (λ / Structural Decomposition of λ)**.**

$$\lambda = \underbrace{\lambda_{redis}} + \underbrace{\lambda_{mobil}} + \underbrace{\lambda_{cohe}} - \underbrace{\lambda_{extr}} - \underbrace{\lambda_{iso}} - \underbrace{\lambda_{trauma}},$$

where

- $\lambda_{redis} \geq 0$——*awaiting*
- $\lambda_{mobil} \geq 0$——
- $\lambda_{cohe} \geq 0$——
- $\lambda_{extr} \geq 0$——
- $\lambda_{iso} \geq 0$——
- $\lambda_{trauma} \geq 0$——

$\lambda < 0$ if and only if

$\lambda < 0$ if and only if the sum of the first three suppressing components is less than the sum of the last three amplifying components.

4.2 / Macro-Aggregation of the Three Conditions

define 4.1 Core

$$\underbrace{(\lambda_{\text{extr}} + \lambda_{\text{trauma}}) - (\lambda_{\text{redis}} + \lambda_{\text{mobil}})}_{\text{ITA}} + \underbrace{(\lambda_{\text{iso}} - \lambda_{\text{cohe}})}_{\text{IPI}} + \underbrace{(\lambda_{\text{iso}} - \lambda_{\text{mobil}})}_{\text{SR}}$$

Three conditions condition domains corresponding to the three core conditions of this paper are identified through this decomposition.

5 (ITA)

6 Condition I: Intergenerational Trauma Amplification (ITA)

6.1 define / Definition and Mechanism

Definition 6.1 (ITA — / Intergenerational Trauma Amplification)**.** $\mathcal{H} = \{H_1, H_2, \ldots, H_K\}$ *each* H_k *define* $\mathbf{T}_k(t) \in \mathbb{R}^d$ d *ITA satisfying*

$$\exists k \in \{1, \ldots, K\} : \frac{d}{dt}\mathbf{T}_k(t) = \underbrace{\mathbf{A}_k \mathbf{T}_k(t)} + \underbrace{\mathbf{B}_k \mathbf{u}_k(t)} - \underbrace{\gamma_k \mathbf{C}_k \mathbf{T}_k(t)}_{\textit{Decays}},$$

where

- $\mathbf{A}_k \in \mathbb{R}^{d \times d}$ *all* > 0*——Decays*
- $\mathbf{u}_k(t) \in \mathbb{R}^m$ *awaiting*
- $\mathbf{C}_k$ $\gamma_k \geq 0$

ITA $\mathbf{A}_k$ $\rho(\mathbf{A}_k) > \gamma_k \|\mathbf{C}_k\|$ *i.e.*

Proposition 6.2 (ITA λ / ITA-λ Linkage)**.** *ITA* λ

$$\lambda \leftarrow \lambda - \eta_{ITA} \cdot \max_k \left(\rho(\mathbf{A}_k) - \gamma_k \|\mathbf{C}_k\| \right),$$

where $\eta_{ITA} > 0$ *ITA* $\rho(\mathbf{A}_k) \gg \gamma_k \|\mathbf{C}_k\|$ *for some* k $\lambda < 0$

6.2 ITA / The ITA Mechanism Chain

λ

Intergenerational trauma amplification drives λ negative through the following causal chain:

1. **Trauma Transmission**
    - **Epigenetic** NR3C1 $\mathbf{A}_k$ $(\mathbf{A}_k)_{ii} \gg \sum_{j \neq i} |(\mathbf{A}_k)_{ij}|$
    - **Behavioral**
    - **Structural**
2. **Extraction Feedback**

    λ_{extr} i.e. $\lambda_{\text{extr}} \uparrow$
3. **Healing Failure** $\gamma_k \|\mathbf{C}_k\|$ Decays—— γ_k

6.3 ITA / ITA Activation Function

Definition 6.3 (ITA / ITA Activation Function)**.** *define ITA* $\Phi_{ITA} : [0,1]^3 \to [0,1]$

$$\Phi_{ITA}(e,p,h) = \frac{1}{1+\exp\left(-\alpha_{ITA}(e \cdot p \cdot h - \tau_{ITA})\right)},$$

where

- $e \in [0,1]$*economic deprivation intensity*
- $p \in [0,1]$*political exclusion degree*
- $h \in [0,1]$*historical trauma persistence, normalized*
- $\alpha_{ITA} > 0$*activation sharpness*
- $\tau_{ITA} \in (0,1)$*activation threshold*

$\Phi_{ITA} \to 1$ *ITA*

7 (IPI)

8 Condition II: Identity Politics Industrialization (IPI)

8.1 define / Definition and Mechanism

Definition 8.1 (IPI — / Identity Politics Industrialization)**.** *there exists M* $\mathcal{I} = \{I_1, \ldots, I_M\}$ *awaitingeach* I_j $\mathcal{E}_j$ $\mathcal{M}_j$ *define*

$$\forall j,\ \underbrace{R_j(t)} = \underbrace{p_j(t)} \cdot \underbrace{q_j(t)} - \underbrace{C_j(\mathbf{s}_j)},$$

where $\mathbf{s}_j \in \mathbb{R}^L$ *IPI Core* λ_{iso} λ_{cohe}

IPI is fundamentally an **industrial logic** applied to identity: identity entrepreneurs compete in a marketplace of grievances, where the unit of production is the "narrative of threat or victimhood," and the unit of consumption is "group-level attention and mobilization." The profit motive ensures that narratives that maximize intergroup hostility are systematically selected.

Proposition 8.2 (IPI λ / Dual Effect of IPI on λ)**.** *IPI*

$$\lambda \leftarrow \lambda - \underbrace{\delta_{iso} \cdot \sum_{j=1}^{M} R_j(t)} - \underbrace{\delta_{cohe} \cdot \sum_{j=1}^{M} \|\mathbf{s}_j\|^2},$$

where $\delta_{iso}, \delta_{cohe} > 0$ *IPI simultaneously* λ $\lambda_{iso} \uparrow$ $\lambda_{cohe} \downarrow$*therefore* λ-

8.2 IPI / IPI Market Dynamics

Definition 8.3 (/ Identity Narrative Competition)**.** *j N_j i*

$$\max_{\mathbf{s}_j^{(i)} \in \mathcal{S}_j} \; p_j^{(i)} \cdot q_j(\mathbf{s}_j^{(i)}, \mathbf{s}_j^{(-i)}) - C_j(\mathbf{s}_j^{(i)}),$$

q_j

$$q_j(\mathbf{s}_j^{(i)}, \mathbf{s}_j^{(-i)}) = \frac{\exp\left(\beta \cdot \mathit{Arousal}(\mathbf{s}_j^{(i)})\right)}{\sum_{k=1}^{N_j} \exp\left(\beta \cdot \mathit{Arousal}(\mathbf{s}_j^{(k)})\right)},$$

where $\mathit{Arousal}(\cdot)$ $\beta > 0$ -

This is a softmax attention market: the more emotionally arousing a narrative is, the larger its audience share. Since threat and victimhood narratives maximize arousal, the market equilibrium selects for maximal intergroup hostility — an emergent, decentralized mechanism driving $\lambda < 0$.

Proposition 8.4 (IPI λ / λ Lower Bound Under IPI Equilibrium)**.** *all $\mathbf{s}^*$ satisfying*

$$\nabla_{\mathbf{s}} \mathit{Arousal}(\mathbf{s}^*) = \beta \cdot \nabla_{\mathbf{s}} C_j(\mathbf{s}^*).$$

λ lower bound

$$\lambda_{IPI} \leq -\delta_{cohe} \cdot M \cdot \|\mathbf{s}^*\|^2.$$

$M \gg 1$i.e. $\lambda < 0$

8.3 IPI / IPI-Information Environment Coupling

Definition 8.5 (/ Algorithmic Amplification Factor)**.** *define*

$$\Gamma_{alg} = \quad \geq 1.$$

$\Gamma_{alg} > 1$ IPI λ

$$\lambda_{IPI}^{\mathit{eff}} = \Gamma_{alg} \cdot \lambda_{IPI}.$$

> *Honest Strike* **[!]** **Strike 1. Honest Strike** IPI multi-armed banditwhere IPI each λ ——

9 (SR)

10 Condition III: Segregation Reinforcement (SR)

10.1 define / Definition and Mechanism

Definition 10.1 (SR — / Segregation Reinforcement)**.** *$\mathcal{G} = (V, E)$ define*

$$\frac{d}{dt} H_{seg}(t) > 0,$$

where H_{seg} *Entropy*

$$H_{seg}(t) = -\sum_{g=1}^{G} p_g(t) \log p_g(t),$$

$p_g(t) = |V_g(t)|/|V| V_g\ t\ g\ SR\ \lambda$

1. ***Physical*** *Schelling Status*
2. ***Informational***
3. ***Marital*** *compression* λ_{mobil}

10.2 SR Schelling / Schelling Dynamics of SR

Definition 10.2 (SR / SR Critical Threshold)**.** *Schelling (1971) define i* $\theta_i \in [0,1]$ *i SR*

$$\theta^* = \inf \{\theta \in [0,1] : \exists \text{ such that } H_{seg} \geq H_{crit}\},$$

where H_{crit} $\mathcal{G}$ *Entropylower bound* θ^* 0.3–0.5

Proposition 10.3 (SR λ / Positive Feedback of SR on λ)**.** H_{seg} H_{crit}

$$\lambda_{iso} \leftarrow \lambda_{iso} + \mu_{SR} \cdot (H_{seg} - H_{crit})^2,$$

$$\lambda_{mobil} \leftarrow \lambda_{mobil} \cdot \exp\left(-\nu_{SR} \cdot (H_{seg} - H_{crit})\right),$$

where $\mu_{SR}, \nu_{SR} > 0$*simultaneously* λ_{iso} λ_{mobil}—— λ

10.3 SR / Irreversibility of SR

Theorem 10.4 (SR theorem / SR Phase Transition Theorem)**.** $\mathcal{G}$ *Entropy* H_{seg} H_{crit} *each* $\rho_g > \rho_{crit}$ *SR*

$$\lim_{t\to\infty} \mathbb{E}[H_{seg}(t)] = \log G,$$

i.e.Status

$$C_{\text{de-seg}}(H_{seg}) \geq C_0 \cdot \exp\left(\alpha \cdot (H_{seg} - H_{crit})\right),$$

where C_0 $\alpha > 0$ -

11 $\lambda < 0$

12 Joint Activation of Three Conditions and $\lambda < 0$

12.1 iff condition / Necessary and Sufficient Condition for Joint Activation

——Divergent Complex

The three conditions do not operate independently—they form a **Divergent Complex** through mutual positive feedbacks:

Definition 12.1 (/ Divergent Complex)**.** $\mathcal{D} = (\Phi_{ITA}, \Phi_{IPI}, \Phi_{SR})$ *Status define*

$$\frac{d}{dt}\begin{pmatrix}\Phi_{ITA}\\ \Phi_{IPI}\\ \Phi_{SR}\end{pmatrix} = \mathbf{J}\begin{pmatrix}\Phi_{ITA}\\ \Phi_{IPI}\\ \Phi_{SR}\end{pmatrix} + ,$$

where $\mathbf{J} \in \mathbb{R}^{3\times 3}$

$$\mathbf{J} = \begin{pmatrix} 0 & j_{12} & j_{13}\\ j_{21} & 0 & j_{23}\\ j_{31} & j_{32} & 0\end{pmatrix}, \quad j_{kl} > 0.$$

Growth

Theorem 12.2 ($\lambda < 0$ iff condition / N&S Condition for $\lambda < 0$)**.** $\lambda < 0$ *iff condition* $\mathbf{J}$ $\rho(\mathbf{J}) > 1$

$$\lambda < 0 \iff \exists k \in \{ITA, IPI, SR\} : \Phi_k > \Phi_k^{crit} \wedge \rho(\mathbf{J}) > 1.$$

$\rho(\mathbf{J}) > 1$

$$(j_{12}j_{23}j_{31} + j_{13}j_{32}j_{21}) + \sqrt{(j_{12}j_{21} + j_{13}j_{31} + j_{23}j_{32})^3} > 1.$$

12.2 / Historical Cases: Condition-Dominance Patterns

Table 1: / Historical Cases of Condition-Dominance Patterns

/ Case	ITA	IPI	SR	/ End State
(1990s)	awaiting		awaiting	
(1994)				
(1850s)		awaiting		
(1980s)		awaiting		
(2016–)				
()				

1 ITA SR simultaneously IPI IPI

Insight from Table 1: When ITA and SR are both strong but IPI is weak, the system can settle into a long-term fragmented steady state (e.g., the Indian caste system). When IPI joins, the steady state is destroyed and the system accelerates toward either extinction or revolution.

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14 Divergent Dynamics

14.1 / Deterministic Divergence

$\lambda < 0$ define 2.1 SDE

Under $\lambda < 0$, the deterministic part of the SDE in Definition 2.1:

Theorem 14.1 (theorem / Exponential Divergence Theorem). $\lambda = -|\lambda| < 0$ $\sigma = 0$

$$\Delta(t) = \Delta_0\, e^{|\lambda| t}.$$

define T_{crit} Δ_{max}

$$T_{crit} = \frac{1}{|\lambda|} \ln\left(\frac{\Delta_{max}}{\Delta_0}\right).$$

Corollary 14.2 (/ Parameter Sensitivity of Divergence Time). T_{crit} $|\lambda|$

$$\frac{d \ln T_{crit}}{d \ln |\lambda|} = -1,$$

awaiting Δ_0

$$\frac{d \ln T_{crit}}{d \ln \Delta_0} = -\frac{1}{\ln(\Delta_{max}/\Delta_0)}.$$

$\Delta_0 \to \Delta_{max}$ *i.e.* Δ_0

14.2 / Stochastic Divergence and Confidence Bounds

$\sigma > 0$ awaiting

When $\sigma > 0$, the inequality path is stochastic. We need probabilistic bounds.

Proposition 14.3 (/ Probabilistic Bounds for Stochastic Divergence). $\lambda = -|\lambda| < 0$ $\sigma > 0$

$$\mathbb{E}[\Delta(t)] = \Delta_0\, e^{|\lambda| t},$$

$$\mathrm{Var}[\Delta(t)] = \Delta_0^2\, e^{2|\lambda| t}\left(e^{\sigma^2 t} - 1\right).$$

$\Delta(t)$ $(1-\alpha)$ *upper boundlower bound*

$$\Delta_{upper}(t) = \Delta_0 \exp\left(|\lambda| t + z_{\alpha/2}\, \sigma \sqrt{t}\right),$$

$$\Delta_{lower}(t) = \Delta_0 \exp\left(|\lambda| t - z_{\alpha/2}\, \sigma \sqrt{t}\right),$$

where $z_{\alpha/2}$ $(1-\alpha/2)$

Proposition 14.4 (/ Distribution of First-Passage Time). *define* $\tau_b = \inf\{t \geq 0 : \Delta(t) \geq b\}$*where* $b > \Delta_0$ $\lambda < 0$ τ_b

$$f_{\tau_b}(t) = \frac{\ln(b/\Delta_0)}{\sigma\sqrt{2\pi t^3}} \exp\left(-\frac{\left(\ln(b/\Delta_0) - |\lambda| t + \frac{\sigma^2}{2} t\right)^2}{2\sigma^2 t}\right).$$

$$\mathbb{E}[\tau_b] = \frac{\ln(b/\Delta_0)}{|\lambda| - \sigma^2/2}, \qquad |\lambda| > \sigma^2/2.$$

14.3 / Divergence Contribution Decomposition

Proposition 14.5 (/ Structural Decomposition of Divergence Rate)**.** $|\lambda|$

$$|\lambda| = \alpha_{ITA}\Phi_{ITA} + \alpha_{IPI}\Phi_{IPI} + \alpha_{SR}\Phi_{SR} - \alpha_0,$$

where $\alpha_{ITA}, \alpha_{IPI}, \alpha_{SR} > 0$ $\alpha_0 \geq 0$ *Decays* $\alpha_0 > \sum \alpha_k \Phi_k$ $\alpha_0 \leq \sum \alpha_k \Phi_k$

15 Status

16 Three Terminal States

$\lambda < 0$ there existsStatus Status

When $\lambda < 0$ persists without intervention, the system necessarily converges to one of three terminal states. The terminal state is determined by the relative activation strength of each condition.

16.1 Status (Fragmentation) — SR

16.2 Terminal State I: Fragmentation — SR-dominant

Definition 16.1 (/ Fragmentation)**.** $\mathcal{S}$ *Statusif and only if*

$$\lim_{t\to\infty} H_{seg}(t) = \log G \quad \wedge \quad \Phi_{SR} > \max(\Phi_{ITA}, \Phi_{IPI}).$$

Status G *each*

$\mathcal{G}$ $\mathbf{A}(t)$ SR

$$\lim_{t\to\infty} \mathbf{A}(t) = \bigoplus_{g=1}^{G} \mathbf{A}_g,$$

i.e. $\mathbf{A}$ spectral gap $\lambda_2(\mathbf{L})$ where $\mathbf{L}$

The mathematical structure of fragmentation: The adjacency matrix $\mathbf{A}(t)$ converges to a block-diagonal form. The spectral gap of the graph Laplacian converges to zero, meaning all cross-group diffusion processes cease entirely.

16.3 Status (Revolution) — IPI

16.4 Terminal State II: Revolution — IPI-dominant

Definition 16.2 (/ Revolution)**.** $\mathcal{S}$ *Statusif and only if*

$$\Phi_{IPI} > \max(\Phi_{ITA}, \Phi_{SR}) \quad \wedge \quad \Delta(t) \ > \Delta_{jump}.$$

Status awaiting awaitingStatus

define $\mathcal{R}(t)$

$$\mathcal{R}(t) = \underbrace{\Phi_{\mathrm{IPI}}(t)} \cdot \underbrace{\Delta(t)} \cdot \underbrace{(1 - \Phi_{\mathrm{SR}}(t))}.$$

$\mathcal{R}(t) \geq \mathcal{R}_{\mathrm{crit}}$ $(1 - \Phi_{\mathrm{SR}})$ if SR IPI

The dynamical condition for revolution: Revolution is triggered when revolutionary potential $\mathcal{R}(t) \geq \mathcal{R}_{\mathrm{crit}}$. The $(1 - \Phi_{\mathrm{SR}})$ term captures a key insight: cross-group alliance is necessary for revolutionary success.

16.5 Status (Extinction) — ITA

16.6 Terminal State III: Extinction — ITA-dominant

Definition 16.3 (/ Extinction)**.** *$\mathcal{S}$ Statusif and only if*

$$\Phi_{ITA} > \max(\Phi_{IPI}, \Phi_{SR}) \quad \wedge \quad \exists g \in \{1, \ldots, G\} : |V_g(t)| \to 0 \text{ as } t \to \infty.$$

Status

g $N_g(t)$ Lotka-Volterra

$$\frac{dN_g}{dt} = r_g N_g \left(1 - \frac{N_g}{K_g}\right) - \underbrace{\eta_g \cdot \Phi_{\mathrm{ITA}} \cdot N_g},$$

where r_g GrowthK_g η_g

$$\eta_g \cdot \Phi_{\mathrm{ITA}} > r_g.$$

The dynamical condition for extinction: Extinction occurs when the trauma-induced carrying-capacity destruction rate $\eta_g \cdot \Phi_{\mathrm{ITA}}$ exceeds the intrinsic growth rate r_g of the target group.

16.7 Status / Phase Diagram of Terminal States

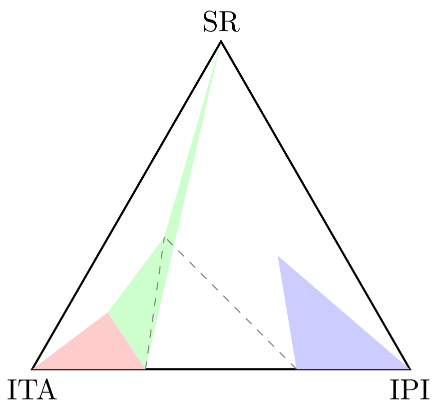

Figure 1: Status / Ternary Phase Diagram of Terminal States by Condition Dominance

1 Status there existswhereawaiting—— simultaneously

Figure 1 shows the distribution of terminal states in the three-condition activation space. Note the **mixed zone** (central region) where all three conditions are near equal strength—this is the most dangerous regime, where the system can simultaneously experience fragmentation and extinction (e.g., the Rwanda case).

17 (CEWI)

18 Five Early Warning Signals and the Composite Early Warning Index (CEWI)

18.1 (ACF)

18.2 Signal I: Rising Autocorrelation (ACF)

Definition 18.1 (-1 / Lag-1 Autocorrelation)**.** $\{\Delta_t\}_{t=1}^{T}$*-1*

$$ACF(1) = \frac{\sum_{t=1}^{T-1}(\Delta_t - \bar{\Delta})(\Delta_{t+1} - \bar{\Delta})}{\sum_{t=1}^{T}(\Delta_t - \bar{\Delta})^2}.$$

$ACF(1)$ $\rho_{crit} \approx 0.7$ *ACF(1)* $\to 1$ *Status*——

In the $\lambda < 0$ regime, ACF(1) increases toward 1 as the system approaches its bifurcation point. This is the classical "critical slowing down" phenomenon: the system's recovery rate from perturbations approaches zero, making it increasingly vulnerable to stochastic shocks.

18.3 (VAR)

18.4 Signal II: Rising Variance (VAR)

Definition 18.2 (/ Rolling Variance)**.** w

$$\sigma_w^2(t) = \frac{1}{w-1}\sum_{\tau=t-w+1}^{t}\left(\Delta_\tau - \bar{\Delta}_w(t)\right)^2.$$

$\sigma_w^2(t)$ *Growth 2*

Under $\lambda < 0$, the variance grows with $\exp(2|\lambda|t + \sigma^2 t)$, making it the most rapidly escalating signal. However, variance can spike for non-catastrophic reasons (election cycles, policy changes), so it must be used in conjunction with ACF.

18.5 (SKEW)

18.6 Signal III: Skewness Shift (SKEW)

Definition 18.3 (awaiting / Skewness of Inequality Distribution)**.**

$$\gamma_1(t) = \frac{\frac{1}{w}\sum_\tau(\Delta_\tau - \bar{\Delta}_w)^3}{\sigma_w^3}.$$

$\gamma_1(t)$ *awaiting——*

A shifting skewness indicates that inequality growth is driven not by broad-based changes but by extreme concentration at the top—the hallmark of the extraction- amplification loop in the ITA condition.

18.7 (SPA)

18.8 Signal IV: Collapse of Spatial Correlation (SPA)

Definition 18.4 (/ Spatial Correlation Index)**.** *R define Moran's I*

$$I(t) = \frac{R}{\sum_{i,j} w_{ij}} \cdot \frac{\sum_{i,j} w_{ij}(\Delta_i - \bar{\Delta})(\Delta_j - \bar{\Delta})}{\sum_i (\Delta_i - \bar{\Delta})^2},$$

where w_{ij} i j $I(t) \to 0$ $I(t) \to -1$ *——SR*

A sharp drop in spatial correlation signals that neighboring regions are diverging in their inequality dynamics, a hallmark of the segregation reinforcement process.

18.9 Mutual Information (MUI)

18.10 Signal V: Decline of Mutual Information (MUI)

Definition 18.5 (Mutual Information / Cross-Group Mutual Information)**.** *for* A B $\mathcal{P}_A$ $\mathcal{P}_B$

$$I(A;B) = \sum_{x\in\mathcal{X}} \sum_{y\in\mathcal{Y}} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}.$$

$I(A;B)$ *DecaysDecays* $I(A;B) \to 0$ A B *—— IPI*

Mutual information decline quantifies the separation of information ecosystems. When $I(A;B) \to 0$, the two groups effectively inhabit different realities—a condition that makes negotiated settlement of grievances structurally impossible.

18.11 (CEWI) / Composite Early Warning Index

——ACF VAR SKEW SPA MUI we propose**CEWI**

Each signal has strengths and weaknesses. Using any single signal risks false alarms or missed detections. We propose the **Composite Early Warning Index (CEWI)**:

Definition 18.6 (CEWI — / Composite Early Warning Index)**.**

$$\mathit{CEWI}(t) = \sum_{k=1}^{5} w_k \cdot z_k(t),$$

where

- $z_k(t)$ k z- $z_k(t) = (x_k(t) - \mu_k^{baseline})/\sigma_k^{baseline}$ *where*

- w_k

$$w_k(t) = \frac{\exp\left(\beta \cdot \mathit{Reliability}_k(t)\right)}{\sum_{j=1}^{5} \exp\left(\beta \cdot \mathit{Reliability}_j(t)\right)},$$

  $\mathit{Reliability}_k(t)$ k t

- $\beta > 0$ $\beta \to \infty$ *max*

Proposition 18.7 (CEWI / Statistical Properties of CEWI)**.** $\lambda \geq 0$*assumption*

$$\mathit{CEWI}(t) \sim \mathcal{N}(0,1) \quad \textit{asymptotic},$$

becauseeach z_k *asymptotic 1*

- ***Watch*** *CEWI* > 1*1-*σ
- ***Warning*** *CEWI* > 2*2-*σ
- ***Alert*** *CEWI* > 3*3-*σ
- ***Critical*** *CEWI* > 4*4-*σ

Proposition 18.8 (CEWI λ / Functional Relationship between CEWI and λ)**.** $\lambda < 0$ *CEWI*

$$\frac{d}{dt}\mathit{CEWI}(t) \approx \gamma_0 + \gamma_1 \cdot |\lambda| \cdot \mathit{CEWI}(t),$$

i.e. CEWI $|\lambda|$ *Growth CEWI* $|\lambda|$

$$|\widehat{\lambda}| \approx \frac{1}{\gamma_1 \cdot \mathit{CEWI}(t)} \cdot \left(\frac{d}{dt}\mathit{CEWI}(t) - \gamma_0\right).$$

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20 Intervention Pathways and Optimal Intervention Timing

20.1 / Phases of Intervention

Intervention effectiveness depends nonlinearly on the timing of entry. We partition the divergence process into three phases:

Definition 20.1 (/ Three Phases of Divergence)**.**

1. ***Phase I — Latent Phase*** *CEWI* $\in [0,1]$$\lambda < 0$ *——i.e.*

   Phase I — Latent Phase: $\lambda < 0$ *has begun but society appears to function normally. Intervention cost is minimal—policy adjustments, dialogue mechanisms, and early redistribution suffice.*

2. ***Phase II — Critical Phase*** *CEWI* $\in (1,3]$ *—— CEWI Growth*

   Phase II — Critical Phase: *Signals become visible; social tension is palpable but not yet erupted. Structural reforms are needed—institutional rebuilding, truth and reconciliation, forced mixing. Cost grows with CEWI.*

3. ***Phase III — Eruptive Phase*** *CEWI* > 3 *vs vs*

   Phase III — Eruptive Phase: *The system has crossed the critical point. Intervention can only influence the type of terminal state (fragmentation vs revolution vs extinction), not prevent collapse.*

20.2 / Condition-Specific Intervention Pathways

20.2.1 ITA / ITA Intervention: Breaking the Trauma Loop

Proposition 20.2 (ITA effectiveness / Effectiveness Condition for ITA Intervention)**.** *ITA* $\rho(\mathbf{A}_k) < \gamma_k \|\mathbf{C}_k\|$

1. ***Institutional Acknowledgment***

   $\mathbf{A}_k$ $\rho(\mathbf{A}_k)$ *30–50%*

2. ***Reparations and Restoration*** $\mathbf{B}_k\mathbf{u}_k$

3. ***Intergenerational Dialogue***

   $\mathbf{A}_k$

ITA - Phase I $> 100:1$ *Phase II awaiting*$10:1$–$50:1$ *Phase III*

20.2.2 IPI / IPI Intervention: De-Industrializing the Identity Market

Proposition 20.3 (IPI / Structural Solution for IPI Intervention)**.** *IPI simultaneously*

1. ***Algorithmic De-Polarization***

$$\mathcal{L}_{new} = \mathcal{L}_{engagement} - \lambda_{cross} \cdot CrossExposure(u),$$

   where $CrossExposure(u)$ u

2. ***Narrative Taxation*** $\tau_{arousal}$

3. ***Public Media Reconstruction***

20.2.3 SR / SR Intervention: Forced Mixing and Boundary Dissolution

Proposition 20.4 (SR - / Dual-Channel SR Intervention)**.** *SR theoremtheorem 10.4* H_{crit}

1. ***Physical Forced Mixing*** *awaiting* $\mathcal{G}$

2. ***Informational Forced Mixing*** *IPI*

3. ***Cross-Group Network Construction***

   Granovetter

20.3 / Optimal Intervention Timing

Theorem 20.5 (theorem / Optimal Intervention Stopping Theorem).

$$V(CEWI) = \max\{i.e.,\ \mathbb{E}[V(CEWI + dCEWI) \mid CEWI]\}.$$

Phase I–II

$$\tau^* = \inf\{t \geq 0 : CEWI(t) \geq CEWI^*\},$$

where $CEWI^*$

$$CEWI^* = \frac{\beta_1}{\beta_1 - 1} \cdot \frac{C_{intervention}}{B_{avoided}},$$

- $C_{intervention}$
- $B_{avoided}$
- $\beta_1 > 1$

***CoreConclusion**$CEWI^* \approx 1.5$–2.0 Phase I Phase II —— CEWI*

Corollary 20.6 (/ Regret Function of Delayed Intervention). τ^* *Growth*

$$Regret(t) = B_{avoided} \cdot [1 - \exp(-|\lambda| \cdot (t - \tau^*))].$$

$|\lambda| \approx 0.05$–0.15/ *5* 22%–53%

21 SCX Axiom System

22 Integration with the SCX Axiom System

λ SCX Axiom Systemtheorem

The λ directionality analysis is not independent of the SCX axiom system but deepens and complements existing theorems:

1. **Thm12theorem** Thm12 $T \propto 1/\Delta^2$ Status Thm12 T Status

   Relation to Thm12 (Civilization Collapse): Thm12 established $T \propto 1/\Delta^2$ but did not differentiate collapse *types*. This paper's three-terminal-state classification supplements Thm12 with a morphology of collapse: the same T can correspond to different terminal states depending on relative condition strengths.

2. **Thm7** IPI SR Thm7 ϵ_{geom}

   Relation to Thm7 (Cross-Domain Partition Preservation): IPI and SR conditions jointly describe the social amplification mechanism of Thm7's geometric component ϵ_{geom}.

3. κ κ awaiting $\lambda < 0$ Why λ CEWI κ $\lambda < 0$ κ λ

   Relation to the κ-suppression framework: κ-suppression explains how $\lambda < 0$ can be *masked*. This paper's early warning signals and CEWI are precisely the tools to detect $\lambda < 0$ while κ-suppression is still effective.

4. **Situs encoding** ITA Situs —— Situs t "" IPI Situs SR Situs Situs

   Relation to the Situs Encoding Framework: The three conditions correspond precisely to distortions of the Situs metric along three orthogonal dimensions: ITA = temporal distortion, IPI = group-dimensional distortion, SR = spatial distortion.

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24 Numerical Simulation and Parameter Calibration

24.1 / Simulation Setup

Euler-Maruyama define 2.1 SDE

To validate theoretical predictions, we discretize the SDE from Definition 2.1:

$$\Delta_{t+\Delta t} = \Delta_t + \lambda\,\Delta_t\,\Delta t + \sigma\,\Delta_t\,\sqrt{\Delta t}\,\epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0,1).$$

Table 2: / Simulation Parameters

/ Parameter	/ Symbol	/ Calibrated Range
awaiting / Initial Inequality	Δ_0	$[0.3, 0.7]$Gini
/ Divergence Rate	$\lvert\lambda\rvert$	$[0.02, 0.15]$ yr^{-1}
/ Volatility	σ	$[0.05, 0.25]$ yr$^{-1/2}$
ITA	α_{ITA}	$[0.005, 0.05]$
IPI	α_{IPI}	$[0.005, 0.08]$
SR	α_{SR}	$[0.002, 0.04]$
CEWI	β	$[1, 5]$

24.2 / Key Simulation Results

1. $\lvert\lambda\rvert = 0.05$, $\Delta_0 = 0.4$, $\Delta_{\max} = 0.8$ $T_{\text{crit}} \approx 14$ 95% $[9, 22]$

   Parameter dependence of divergence time: Under baseline parameters, median $T_{\text{crit}} \approx 14$ years with 95% CI $[9, 22]$ years.

2. **CEWI** CEWI 5–8 CEWI > 2 2–4 CEWI > 3

   Lead time of CEWI: Yellow warning (CEWI > 2) leads collapse by 5–8 years on average; red alert (CEWI > 3) by 2–4 years.

3. Phase I CEWI ≈ 1.5 $> 90\%$ $\lambda \geq 0$ Phase II CEWI ≈ 2.5 40%–60% Phase III CEWI > 3 $< 10\%$

   Timing sensitivity of intervention: Phase I-end intervention has $> 90\%$ probability of restoring $\lambda \geq 0$. Phase II-mid intervention drops to 40%–60%. Phase III intervention $< 10\%$ (limited to terminal-state type conversion).

25

26 Discussion

26.1 / Contemporary Applicability of the Three Conditions

2026simultaneously

In the global society of 2026, all three conditions are simultaneously activated at historically unprecedented intensities:

- **ITA** ITA —— Trauma Inflation

  ITA: Legacies of colonialism, slavery, and genocide are reactivated in the social media era. Crucially, modern ITA is no longer confined to historically victimized groups—social media commoditizes trauma narratives, enabling a phenomenon of **Trauma Inflation**.

- **IPI** 2026 \$2500 where $> 60\%$ IPI $R_j(t)$

  IPI: Recommendation algorithms have elevated identity politics industrialization to industrial scale. In 2026, global social media ad revenue exceeds \$250B, with identity-narrative-driven engagement accounting for $> 60\%$.

- **SR** + awaiting + =

  SR: Algorithmic filter bubbles + residential segregation driven by economic inequality + remote work reducing physical contact = a perfect storm for segregation reinforcement.

26.2 CEWI / Operational Recommendations for CEWI

1. **stability** SSMASocial Stability Monitoring Agency CEWI Core stability **Institutional Design:** Establish independent Social Stability Monitoring Agencies (SSMAs) with CEWI as the core metric.

2. CEWI awaitingGini

   Mutual Information **Data Infrastructure:** CEWI computation requires high-quality data on inequality, spatial segregation, and information ecosystem metrics.

3. due to IPI CEWI CEWI **International Coordination:** Since IPI's information channels are transnational, single-country CEWI monitoring may miss narrative attacks originating abroad. International CEWI cooperation is needed.

26.3 Honest Strike / Honest Critique: Reflexivity Risk of Early Warning

Honest Strike [!] **Strike 2. Honest StrikeReflexivity Risk**

This paper proposes CEWI CEWI λ

- CEWI $\rightarrow$ $\rightarrow$ $\rightarrow$ awaiting $\rightarrow$ CEWI ——
- CEWI $\rightarrow$ $\rightarrow$ IPI ——

CEWI —— i.e. CEWI ——“”

27 Conclusion

28 Conclusion and Outlook

28.1 Core / Core Contributions

This paper systematically formalizes SCX Equality Principle $\lambda < 0$ Core

This paper systematically formalizes the $\lambda < 0$ divergent regime within the SCX Equality Principle framework. Core contributions include:

1. $\lambda < 0$ ITAIPI SR eachdefine——
2. **StatusClassification:** $\Delta(t) = \Delta_0 e^{|\lambda|t}$ Status
3. **CEWI**
4. Phase I Phase II Conclusion

28.2 Open Problem / Open Problems

1. **CEWI** CEWI 1990–19951990–1994 2010–2011awaiting

   Empirical validation of CEWI: Retrospective testing on historical collapse cases (Former Yugoslavia, Rwanda, Syria) is needed.

2. λ awaiting Mutual Information λ Status

   Real-time λ estimation: How to estimate λ and its components from high-frequency data? Real-time state-space models are needed.

3. CEWI Causal Inference

   RCTs of CEWI-guided interventions: Effects of interventions such as algorithmic depolarization and forced mixing need rigorous causal inference.

4. $\lambda < 0$ λ

   Cross-scale coupling: This paper focuses on national-scale $\lambda < 0$. How do community, city, and global-scale λ dynamics couple?

5. **AI λ** AI simultaneously ITA IPI SR AI λ

   Net effect of AI on λ: Generative AI simultaneously amplifies ITA, IPI, and SR. Its net effect on λ components requires systematic modeling.

A AppendixACEWI

B Appendix A: Bayesian Update of CEWI Weights

define 18.6 CEWI

In Definition 18.6, CEWI weights are time-varying. The update rule:

Proposition B.1 (CEWI / CEWI Weight Update)**.** *$\mathcal{D}_t = \{(x_k(\tau), y(\tau))\}_{\tau=1}^{t}$ t where $y(\tau) \in \{0,1\}$ $y = 1$ t Δt k reliabilitydefine*

$$Reliability_k(t) = \frac{TP_k(t) + TN_k(t)}{TP_k(t) + TN_k(t) + FP_k(t) + FN_k(t) + \epsilon},$$

where TP, TN, FP, FN False PositiveFalse Negative $\epsilon > 0$ reliability softmax

C AppendixB

D Appendix B: Differential Equation System of the Three Conditions

define 12.1

The full dynamics of the Divergent Complex (Definition 12.1) is governed by the coupled nonlinear system:

$$\frac{d\Phi_{\text{ITA}}}{dt} = -\gamma_{\text{ITA}}\Phi_{\text{ITA}} + \frac{\beta_{\text{ITA}}}{1 + \exp(-\alpha_{\text{ITA}}(c_{12}\Phi_{\text{IPI}} + c_{13}\Phi_{\text{SR}} - \tau_{\text{ITA}}))}, \tag{1}$$

$$\frac{d\Phi_{\text{IPI}}}{dt} = -\gamma_{\text{IPI}}\Phi_{\text{IPI}} + \frac{\beta_{\text{IPI}}}{1 + \exp(-\alpha_{\text{IPI}}(c_{21}\Phi_{\text{ITA}} + c_{23}\Phi_{\text{SR}} - \tau_{\text{IPI}}))}, \tag{2}$$

$$\frac{d\Phi_{\text{SR}}}{dt} = -\gamma_{\text{SR}}\Phi_{\text{SR}} + \frac{\beta_{\text{SR}}}{1 + \exp(-\alpha_{\text{SR}}(c_{31}\Phi_{\text{ITA}} + c_{32}\Phi_{\text{IPI}} - \tau_{\text{SR}}))}, \tag{3}$$

where γ_k Decaysβ_k c_{kl} τ_k

E AppendixC

F Appendix C: Dialogue with Contemporary Theories

F.1 C.1 / Relation to Collapse Archaeology

Tainter (1988) λ Perspective ITAIPI SR —— Why

Tainter's (1988) *Collapse of Complex Societies* theorized collapse as driven by diminishing marginal returns. The λ framework provides a complementary perspective: collapse is not merely a resource-efficiency problem but a social-structural condition problem. ITA, IPI, and SR can independently drive collapse even under conditions of resource abundance—explaining why some resource-rich civilizations (e.g., the Classic Maya) collapsed before reaching diminishing marginal returns.

F.2 C.2 Piketty / Relation to Piketty's Capital Dynamics

Piketty (2014) Coreawaiting $r > g$Growthawaiting λ awaiting $r > g$ Δ $\lambda < 0$ Δ $\lambda < 0$ i.e. $r = g$Piketty awaiting

Piketty's (2014) core inequality $r > g$ describes the *material driver* of inequality. The λ framework provides the *social-structural driver*. The two are complementary, not substitutes: $r > g$ determines the *magnitude* of Δ, while $\lambda < 0$ determines its *direction* (self-correcting vs. self-amplifying). Under $\lambda < 0$, even if $r = g$ (Piketty's equilibrium condition), structural self-amplification of inequality can still lead to collapse.

F.3 C.3 / Relation to Critical Transition Theory

Scheffer et al. (2009, 2012) Critical Transitions ——flickering

(1) CEWI —— (2) ——SPAMutual InformationMUI

Scheffer et al.'s (2009, 2012) Critical Transition theory identified generic early warning signals near bifurcation points—critical slowing down, rising variance, and flickering. This paper's five early warning signals directly build on that foundation but make two extensions: (1) synthesis from individual signals to the composite CEWI index—resolving the problem of conflicting individual signals; (2) specialization from generic systems to social systems—adding spatial correlation collapse (SPA) and mutual information decline (MUI) as social-system-specific signals.

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